Worksheet 7: Exoplanets and Habitable Worlds

Covers Lectures 13 and 14: transit and radial-velocity detection, atmospheric characterisation, the habitable zone, direct imaging, the Drake equation

Planetary Systems (WBAS002-05) • Kapteyn Astronomical Institute, University of Groningen

This worksheet is ungraded practice, with the same shape and level as the final exam. Plan up to 2 hours for a serious pass. Work each problem through on paper before you open the solutions, and show your algebra: the exam asks for the steps, not only the number. Some parts state an intermediate result ("show that …"); use it to check your work, and to continue if a step resists. Carry at least four significant figures through the intermediate steps (the worked solutions print five) and round only at the end. All parts are analytical; no computer is needed.

Constants and data

$G = 6.674 \times 10^{-11}$ m^3 kg^{-1} s^{-2}, $\sigma = 5.670 \times 10^{-8}$ W m^{-2} K^{-4}, $k_B = 1.381 \times 10^{-23}$ J K^{-1}, $u = 1.661 \times 10^{-27}$ kg

1 AU = 1.496×10^{11} m, 1 pc = 3.086×10^{16} m, 1 yr = 3.156×10^7 s, 1 d = 86,400 s, 1 rad = 206,265 arcsec

Sun: $M_\odot = 1.989 \times 10^{30}$ kg, $L_\odot = 3.828 \times 10^{26}$ W, $R_\odot = 6.957 \times 10^8$ m. Earth: $M_\oplus = 5.972 \times 10^{24}$ kg, $R_\oplus = 6.371 \times 10^6$ m, Bond albedo $A_B = 0.3$. Jupiter: $M_J = 1.898 \times 10^{27}$ kg

Hot Jupiter (a 51 Peg b analogue): $R_p = 8.6 \times 10^7$ m, $m_p \sin i = 0.5\,M_J$, $P = 4.23$ d, $a = 0.05$ AU, Sun-like host. Terminator: $T = 1500$ K, $\mu = 2.3$, $g = 8.6$ m s^{-2}, $n_H = 5$. K2-18 b: $m_p = 8.6\,M_\oplus$, $R_p = 2.6\,R_\oplus$

Rocky planet with a CO_2 atmosphere: $T = 300$ K, $\mu = 44$, $g = 9.81$ m s^{-2}. Spectrograph precision: ELODIE ~ 10 m s^{-1}; ESPRESSO 0.10 m s^{-1} (photon limit), ~ 0.5 m s^{-1} (long-term repeatability)

Kopparapu habitable-zone limits (Sun-like stars): runaway greenhouse $S_{\mathrm{in,eff}} = 1.06$, maximum CO_2 greenhouse $S_{\mathrm{out,eff}} = 0.35$. K dwarf: $L = 0.3\,L_\odot$, $M = 0.7\,M_\odot$. M dwarf (TRAPPIST-1-like): $L = 5 \times 10^{-4}\,L_\odot$. $t_{\mathrm{MS},\odot} = 10$ Gyr

Reflected-light contrast at visible wavelengths: ~ 10^{-9} for a Jupiter analogue at 5 AU, ~ 10^{-10} for an Earth analogue; the Roman coronagraph reaches 10^{-8} to 10^{-9}. Data as in the Lecture 13 and 14 notes.

Problem 1 Transit depths and the geometric transit probability

The transit method measures the fraction of starlight a planet blocks, $\delta = (R_p/R_\star)^2$, but only for the small fraction of orbits aligned with our line of sight: a randomly oriented orbit transits with probability $p \approx R_\star/a$. These two numbers together explain what the transit catalogue contains, and what it misses.

(a) Compute the transit depth of the hot Jupiter in front of its Sun-like host.

(b) Repeat for an Earth analogue transiting a Sun-like star, and express the result in parts per million.

(c) Compute the geometric transit probability for the Earth analogue at 1 AU and for the hot Jupiter at 0.05 AU, and the ratio of the two.

(d) The transit catalogue is dominated by short-period planets and by planets of M dwarfs. Using your results from parts (a) to (c), explain in two or three sentences why this does not mean that such planets are intrinsically the most common.

Problem 2 Radial-velocity masses and the density of K2-18 b

A planet and its star orbit their common centre of mass, so the star moves with a radial-velocity semi-amplitude

$$K_\star = \left(\frac{2\pi G}{P}\right)^{1/3} \frac{m_p \sin i}{M_\star^{2/3}}$$

for $m_p \ll M_\star$ and a circular orbit. Combining an RV mass with a transit radius gives the bulk density, the first handle on what a planet is made of.

(a) Compute $K_\star$ for the 51 Peg b analogue ($m_p \sin i = 0.5\,M_J$, $P = 4.23$ d) around a $1\,M_\odot$ star, and compare it with the $\sim 10\ \mathrm{m\,s^{-1}}$ precision of ELODIE, the spectrograph that found 51 Peg b.

(b) Compute $K_\star$ for an Earth analogue ($m_p = M_\oplus$, $P = 1$ yr) around a $1\,M_\odot$ star, and compare it with the ESPRESSO photon-noise limit and long-term repeatability.

(c) K2-18 b has $m_p = 8.6\,M_\oplus$ from radial velocities and $R_p = 2.6\,R_\oplus$ from transits. Compute its bulk density and compare it with Earth's mean density of $5514\ \mathrm{kg\,m^{-3}}$.

(d) Explain in two or three sentences why a radial-velocity detection alone gives only a minimum mass, and why the combination with a transit removes the ambiguity.

Problem 3 Atmospheric scale heights and transmission spectroscopy

During a transit, a small fraction of the starlight filters through the planet's terminator, and absorbing gases make the planet look larger at their wavelengths. The size of the effect is set by the atmospheric scale height $H = k_B T/(\mu\, u\, g)$, and the fractional change of the transit depth across a strong line is $\Delta\delta/\delta \approx 2 n_H H/R_p$ for n_H scale heights of modulation.

(a) Compute the scale height of the hot Jupiter's terminator ($T = 1500$ K, $\mu = 2.3$, $g = 8.6\ \mathrm{m\,s^{-2}}$).

(b) Compute the scale height of the rocky planet's CO_2 atmosphere ($T = 300$ K, $\mu = 44$, $g = 9.81\ \mathrm{m\,s^{-2}}$), and the ratio of the two scale heights.

(c) For the hot Jupiter, compute the fractional modulation $\Delta\delta/\delta = 2 n_H H/R_p$ with $n_H = 5$, then the absolute change in transit depth $\Delta\delta$ using the hot-Jupiter transit depth $\delta = 1.528 \times 10^{-2}$ of Problem 1(a), in ppm.

(d) JWST observed the rocky planet TRAPPIST-1 b in transmission and found a featureless spectrum, and its $15\,\mu$m secondary eclipse gave a dayside brightness temperature of ≈ 503 K, matching the 508 K bare-rock prediction. Explain in two or three sentences why the flat transmission spectrum alone could not settle whether the planet has an atmosphere, and what the eclipse measurement added.

Problem 4 Equilibrium temperature and the habitable zone

A planet's equilibrium temperature follows from the balance of absorbed starlight and emitted thermal radiation,

$$T_{\text{eq}} = \left[\frac{L_\star(1 - A_{\text{B}})}{16\pi\sigma\epsilon d^2}\right]^{1/4},$$

and the habitable zone is the strip where the effective stellar flux S_{eff} lies between the runaway-greenhouse and maximum-CO_2-greenhouse limits, at orbital distances

$$d = (1\ \text{AU})\sqrt{\frac{L_\star/L_\odot}{S_{\text{eff}}}}.$$

(a) Compute Earth's equilibrium temperature ($A_{\text{B}} = 0.3$, $\epsilon = 1$, $d = 1$ AU), and reconcile it with the observed mean surface temperature of 288 K.

(b) Compute the inner and outer habitable-zone distances for the Sun using $S_{\text{in,eff}} = 1.06$ and $S_{\text{out,eff}} = 0.35$, then the S_{eff} received by Venus (0.72 AU) and Mars (1.52 AU), and place both planets relative to the zone.

(c) Compute the habitable-zone boundaries for the K dwarf ($L = 0.3\,L_\odot$) and the M dwarf ($L = 5 \times 10^{-4}\,L_\odot$), then the main-sequence lifetime of the K dwarf ($M = 0.7\,M_\odot$) from $t_{\text{MS}} \approx t_{\text{MS},\odot}\,(M_\star/M_\odot)^{-2.5}$.

(d) M dwarfs offer the deepest transits, the largest transit probabilities in the habitable zone, and the longest lifetimes, yet they are not automatically the best hosts for life. Name the main physical drawbacks of an M-dwarf habitable zone in two or three sentences.

Problem 5 Direct imaging and the Drake equation

Direct imaging separates the planet's photons from the star's on the detector. By the small-angle formula the separation on the sky is $\theta = a/d$, which by the definition of the parsec is $\theta[\text{arcsec}] = a[\text{AU}]/d[\text{pc}]$. The final problem turns from detection to the question the whole field serves: how many inhabited worlds are out there?

(a) Compute the angular separation of an Earth analogue ($a = 1$ AU) and a Jupiter analogue ($a = 5$ AU) around a Sun-like star at $d = 10$ pc, in arcseconds.

(b) Using the habitable-zone boundaries from Problem 4 (0.9713 to 1.690 AU for the Sun, 0.02172 to 0.03780 AU for the M dwarf), compute the angular extent of the Sun's habitable zone and of the M dwarf's habitable zone, both at 10 pc.

(c) The Drake equation factorises the number of communicating civilisations as

$$N = R_\star f_p n_e f_l f_i f_c L,$$

and only the first three factors are measured. Model the four unconstrained factors as independent and log-uniform, each with $\log_{10}$ distributed uniformly on $[-10, 0]$, and take the measured prefactor as $\log_{10}(R_\star f_p n_e) \approx 0$. Compute the mean and standard deviation of $\log_{10} N$, and the number of standard deviations separating the mean from $N = 1$.

(d) The lectures presented the Drake equation as a framing tool, not an estimator. Give the main reasons in two or three sentences.